Abderrahmane **Ait gueni ssaid**

Machine Learning Engineer · M.S. Computer science

Grenoble, France

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Profile

Experienced ML Engineer with an M.S. in Data Science & AI (Grenoble-INP, Ensimag). Research and industry experience on interpretable ML, knowledge editing in LLMs, time-series forecasting, and applying ML to medical imaging. Comfortable across the ML lifecycle: research, prototyping, engineering and deployment. Proficient in Python, C/C++, CUDA, PyTorch, and production tooling (Docker, FastAPI, MLflow, Ray).

Skills

Languages
English: C2 - EF SET
French: C1
German: A1
Arabic: Native
Kabyle: Native

Programming
Python
C/C++
Cuda
Java
SQL
MPI
OpenMP

Packages
Pytorch
scikit-learn
SciPy
Pandas
Transformers
LangChain
Ollama
Elasticsearch
Matplotlib
OpenCV
FastAPI
MLflow
Ray
pytest

Tools
Linux
Shell
Docker
UV
Anaconda
Git
LaTeX

Publications

Factual Knowledge Assessment For Language Models Using Distractors — COLING 2025, ACL
H. A. Khodja, **A. Ait guenissaid**, F. Béchet, Q. Brabant, A. Nasr and G. Lecorvé.

Internships

ML Engineer — Orange - Lannion, FR
Evaluation of knowledge editing in large language models (LLMs). — Mar. 2024 - Aug. 2024
- Co-defined a **Knowledge Measure** to evaluate atomic knowledge editing of LLMs, incorporated temporality into the metric.
- Validated the measure across multiple LLMs and baseline random LMs; ran extensive controlled experiments and analysis.

ML Engineer — Grenoble Informatics Laboratory, FR (REMOTE)
Integrating information retrieval constraints into deep neural networks. — May 2023 - July 2023
- Reformulated IR constraints within a neural framework and developed the mathematical foundations for model integration.

ML Engineer — Grenoble Institute of Neurosciences, FR
Multi-parametric data integration for imaging biomarker development in epilepsy. — Feb. 2022 - Aug. 2022
- Implemented clustering (k-means, GMM) for multi-parametric imaging data; evaluated models with AIC/BIC.

ML Engineer — Grenoble Informatics Laboratory, FR
Data augmentation with GANs for semi-supervised classification on images. — June 2021 - July 2021
- Implemented semi-supervised models and experimented with GANs, SVMs and Random Forests for microscopic image tasks.

Education

M.S. in Data Science & Artificial Intelligence — Grenoble INP - Ensimag, FR
2nd year GPA: 14/20 (High Honors). 1st year GPA: 13/20 (Top 30% of class). — Aug. 2024

Magistère in Computer Science — Université Grenoble Alpes, FR
University diploma - GPA: 14/20 (High Honors). — Aug. 2024

B.S. in Computer Science — Université Grenoble Alpes, FR
Aug. 2021

Certifications

Mathematics for Machine Learning: Linear Algebra — Imperial College London (Coursera), 2023

Projects

2025

- **Court Case Analytics Pipeline**: Built an automated pipeline to download and process 700k legal case records using Polars; indexed data in Elasticsearch to enable full-text search, scalable storage using DuckDB, and real-time trend analysis.
- **Retail Forecasting Microservice**: Built a production-aware pipeline using Prophet for modeling, Ray for parallel training, MLflow Registry for model versioning and monitoring, FastAPI for low-latency inference, and Docker/Kubernetes for scalable deployment.
- **LangChain Chat with Your Data**, deeplearning.ai
- **Made With ML** Combined ML with software engineering principles for production-ready ML applications.

2024

- **HomeLab**: A home server running multiple VMs and docker containers nodes on a Proxmox hypervisor.
- **Finetuning LLMs**, deeplearning.ai.
- **Pretraining LLMs**, deeplearning.ai.
- **Let's reproduce GPT-2 (124M)**, Andrej Karpathy Tutorial.
- **GPU-Accelerated Multilayer Perceptron from scratch** for MNIST Digits Classification.

2023

- **ML Algorithms from scratch**, Perceptron, Gradient descent, K-means, MLP.
- **Let's build GPT: from scratch, in code, spelled out**, Andrej Karpathy Tutorial.
- **SAT Solver**, Automated Planning and Problem Solving using PDDL4J and Sat4J.
- **Distributed Messaging System using Virtual Overlay Ring Network**, Designed and implemented using Java and RabbitMQ, highlighting expertise in distributed systems and routing algorithms.
- **System Programming - NachOS**. Collaborated in a team to implement processes, threads and synchronization features in the NachOS, gaining experience in OS design, implementation and multithreading.

2022

- **Balloon Breaker**: Developed a ROS-based C++ pipeline, coding the core contour-based balloon detection algorithm and integrating real-time tracking and control nodes; validated across multi-color scenarios.
- **Internships web scraping** Developed an automated application to extract and organize internship proposals from the faculty website into a CSV format. Python, Pandas, BeautifulSoup.

Work Experience

Student Tutor (self-employed) — Grenoble, FR

Private tutoring in maths, physics, chemistry and computer science. — Sept. 2019 - Present

- Taught students of varied ages; developed pedagogy and adaptable lesson plans.
- Skills: communication, patience, time management, problem solving.

Education Assistant — Grenoble High School, FR

Residential support and student supervision. — Sept. 2021 - Aug. 2023

- Facilitated student growth through supervision and mentorship; strengthened leadership and mediation skills.

Student Tutor — Solution Bilingue, FR

Taught maths, physics, chemistry and computer science, to pupils across school levels. — Dec. 2018 - June 2019

- Skills: teaching, communication, patience, problem-solving, adaptability, time management.

Volunteering

Technical Support, Bread Collection, Local Cleaning — Local community associations

- Maintained AV and surveillance systems; provided technical assistance for events.
- Coordinated bread collection for charity distribution.
- Participated in local community cleaning and maintenance activities.